Race, Gender, and the Visual Culture of Domestic Labor: An Interactive Digital Archive of Tradecards and Postcards from the age of New Imperialism

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"Race, Gender, and the Visual Culture of Domestic Labor" is a freely and publicly accessible digital humanities project that brings together an extensive archive of tradecards and postcards depicting domestic labor, from the 1870s to the 1940s. We look at the evolution and representation of various domestic laborer stereotypes such as the South Asian "Ayah", the South-East Asian "Baboe", the African American "Mammy", the Irish "Bridget", the Mexican "Criada", the Algerian "Harem-Servant", the Chinese "Laundry-Man", and Asian and African "House-Boys". The project not only historicizes these domestic servant figures in the specific regional histories of the United States, the British, French, and Dutch colonial empires, but also draws trans-imperial connections in colonial caregiving and visual cultures in the period of New Imperialism (1870s to 1930s), which marked an era of unprecedented Euro-American colonial expansionism in Africa, Asia, and Latin America. We show that despite local differences, there were similarities in the way black and brown domestic workers were sentimentalized, exoticized, animalized, hysersexualized or dexesualized, in the period of race-based segregation, xenophobia, and the naturalization of gendered and racialized domestic labor for white families across the world. By digitizing, analyzing, and exhibiting the mass-produced visual ephemera, the project provides a nuanced historical context for their popularity during the period of British and French New Imperialism, American Jim Crow racism, Asian Exclusion, and concurrent movements for women's suffrage and labor rights. The project also explores the contemporary legacies of these imperial visuals in perpetuating racial and gender hierarchies.

This digital humanities project is an interdisciplinary collaborative effort between history and computer science faculty and students at The College of New Jersey and Khoury College of Computing Sciences at Northeastern University. The work-in-progress version of the project can be found here: https://visualdomesticlaborhistory.khoury.northeastern.edu/ A previous version of the project can be found here: https://empirehistorycards.cs.stonybrook.edu/ .

The publicly accessible website is being designed using value-sensitive design principles (Friedman 1996) to ensure an interactive, educative, ethical, and reflective experience. Given the racist and sexist nature of many images, the website incorporates layered and ethically conscious design strategies to contextualize these dehumanizing depictions. Through innovative visualization techniques and interactive features, this web archive offers a dynamic alternative to traditional digital archives, fostering critical engagement and promoting anti-racism and feminism in public and educational contexts.

The website prioritizes interactivity to enhance user engagement and learning. The cards are presented through multiple visualization methods, offering users diverse pathways to explore the archive. One visualization maps the circulation of these cards geographically, using historical and contemporary maps to provide spatial insights into their dissemination and impact (Moyroud, Nicolas / Frédéric Portet 2018: 1). Another categorizes the cards by broad themes, making them searchable and presenting patterns in the stereotypes represented. These interactive features allow users to critically analyze how global systems of race, gender, and empire operated through everyday visual culture. Although visual ephemera is largely marginalized in academic studies of imperialism, our digital humanities project argues that mass-produced everyday objects played an important role in upholding the hierarchies of race, gender, and empire.

Understanding the artifacts' historical significance and harmful legacy necessitated a design approach that foregrounds ethical considerations. Many of the tradecards and postcards contain overtly racist and sexist imagery. To mitigate the risk of misinterpretation or uncritical consumption, we implemented several visual effects (e.g., blurring and masking) for explicitly problematic images. Furthermore, users of the website are re-

quired to engage with accompanying interpretive text in transcripts before the images are revealed. This layered design ensures that users approach the visuals with informed context, creating space for meaningful reflection.

Moreover, the platform includes a unique feature that enables users to rotate and view all sides of the tradecards and postcards. The backsides of these artifacts often contain invaluable historical information—such as advertisements, personal messages, and printing details—that provide further context to their creation and use. This holistic approach underscores the importance of viewing these items as multidimensional historical documents, not merely as static images.

The web platform has been developed as a client-server architecture (Hanson 2004), which means that the system is composed of two broad subsystems, a client system and a server system. The client system is responsible for rendering and presenting the content. We used React (Boduch 2019) as the primary rendering engine for displaying the content. Further, we use several visualization libraries (e.g., leaflet, material UI, QGIS, etc.) to create the visualizations for the displayed content. The server system encapsulates the logic to analyze, store, and generate the content used for the presentation. This content includes the postcard and tradecard images, their metadata (e.g., location, themes, etc.), and their corresponding transcripts, which contain interpretive texts for the images. We used Node.js for the server system and MongoDB, a non-relational database to store the raw data (Low et al. 2014: 283).

A client-server architecture is particularly well suited in this context for several reasons. First, it promotes separation of concerns (Mili et al. 2004: 75). The logic to render and present the content is kept separate in client components from the logic to analyze, store, and generate the content, which is kept in server components. This is beneficial from a software maintenance perspective as changes to the client and server can be made independently without affecting each other. Further, the client and server components can use different technologies and programming languages. This is essential as the technical requirements and constraints needed to create visualizations are different from those needed to store and analyze the raw data in the historical artifacts. Second, it promotes loose coupling between the system's components. Hence, if there is a need to change the way we present the data in the future, we can replace the current client components with different client components without changing the way the raw data is stored and analyzed. Third, the chosen architecture enables the scalability and performance needs of the website. Using separate server components means that we can scale the server easily without affecting the client components if traffic to the website increases. This is possible due to the use of Node.js and MongoDB, the underlying technology, used to build the server components. Both Node.js and MongoDB enable asynchronous programming and horizontal scaling, techniques widely used to improve the performance and responsiveness of software systems.

In addition to its scholarly contributions, the project seeks to serve as a teaching tool for high school and college educators. By presenting these artifacts in an accessible and interactive format, the website encourages students to critically examine the histories of racism, xenophobia, and patriarchy during the era of global empires. The thematic visualizations, along with interpretive essays and metadata, allow educators to design lessons that connect past stereotypes to their enduring legacies in contemporary culture.

Furthermore, through this work, we aim to stimulate broader conversations about the role of visual culture in perpetuating racialized and gendered hierarchies. By making this collection freely accessible, the website invites users to grapple with uncomfortable histories and reflect on how visual media has been used to reinforce oppression. This engagement is framed not as a passive viewing experience but as an active, educative encounter that fosters critical thinking and historical empathy.

Unlike traditional digital archives, this project uses a layered, user-centric model that balances accessibility with critical reflection. The deliberate inclusion of interactive features transforms the archive from a repository of static images into a dynamic platform for analysis and learning.

The thematic and spatial visualizations, combined with the ability to explore all artifacts holistically, embody an innovative approach to digital archiving. These features not only enhance the users' experience but also enable them to understand these materials within their historical, cultural, and geographical contexts. The project highlights and emphasizes the role of ethical design in engaging with sensitive historical materials.

In conclusion, by digitizing and critically presenting an archive of tradecards and postcards that perpetuated racialized and gendered stereotypes, our project offers an educative platform for public and academic audiences alike. Through its user-centered design, client-server architecture, interactive features, and thematic and spatial visualizations, the website invites users to engage with these artifacts in thoughtful and informed ways. By fostering conversations about the contemporary legacies of these historical ma-

terials, the project contributes to broader efforts to promote anti-racism and feminism. It challenges users to reflect on the enduring impact of visual culture in shaping societal hierarchies and offers a model for using digital tools to confront difficult histories with care and critical insight.

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